

Spring Rest HandsOn

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.3</version>
  </parent>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
    </dependency>

    <dependency>
      <groupId>com.fasterxml.jackson.dataformat</groupId>
      <artifactId>jackson-dataformat-xml</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-test</artifactId>
      <scope>test</scope>
    </dependency>

  </dependencies>
```

</project>

SpringLearnApplication.java

```
package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {

    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
    }
}
```

Country.java

```
package com.cognizant.springlearn.model;

public class Country {

    private String code;
    private String name;

    public Country() {
    }

    public Country(String code, String name) {
        this.code = code;
        this.name = name;
    }

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }
}
```

```
    public String getName() {  
        return name;  
    }  
  
    public void setName(String name) {  
        this.name = name;  
    }  
}
```

HelloController.java

```
package com.cognizant.springlearn.controller;  
  
import org.slf4j.Logger;  
import org.slf4j.LoggerFactory;  
import org.springframework.web.bind.annotation.GetMapping;  
import org.springframework.web.bind.annotation.RestController;  
  
@RestController  
public class HelloController {  
  
    private static final Logger LOGGER =  
        LoggerFactory.getLogger(HelloController.class);  
  
    @GetMapping("/hello")  
    public String sayHello() {  
  
        LOGGER.info("START");  
  
        String result = "Hello World!!";  
  
        LOGGER.info("END");  
  
        return result;  
    }  
}
```

CountryService.java

```
package com.cognizant.springlearn.service;  
  
import java.util.ArrayList;  
import java.util.List;
```

```

import org.springframework.stereotype.Service;

import com.cognizant.springlearn.exception.CountryNotFoundException;
import com.cognizant.springlearn.model.Country;

@Service
public class CountryService {

    public Country getCountryIndia() {
        return new Country("IN", "India");
    }

    public List<Country> getAllCountries() {

        List<Country> list = new ArrayList<>();

        list.add(new Country("IN", "India"));
        list.add(new Country("US", "United States"));
        list.add(new Country("JP", "Japan"));
        list.add(new Country("DE", "Germany"));

        return list;
    }

    public Country getCountry(String code)
        throws CountryNotFoundException {

        List<Country> countries = getAllCountries();

        return countries.stream()
            .filter(c -> c.getCode()
                .equalsIgnoreCase(code))
            .findFirst()
            .orElseThrow(() ->
                new CountryNotFoundException());
    }
}

```

CountryNotFoundException.java

```
package com.cognizant.springlearn.exception;

import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.ResponseStatus;

@ResponseStatus(
    value = HttpStatus.NOT_FOUND,
    reason = "Country not found")
public class CountryNotFoundException
    extends Exception {

    private static final long serialVersionUID = 1L;
}
```

CountryController.java

```
package com.cognizant.springlearn.controller;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;

import com.cognizant.springlearn.exception.CountryNotFoundException;
import com.cognizant.springlearn.model.Country;
import com.cognizant.springlearn.service.CountryService;

@RestController
public class CountryController {

    @Autowired
    private CountryService service;

    @GetMapping("/country")
    public Country getCountryIndia() {

        return service.getCountryIndia();
    }

    @GetMapping("/countries")
```

```

    public List<Country> getAllCountries() {

        return service.getAllCountries();
    }

    @GetMapping("/country/{code}")
    public Country getCountry(
        @PathVariable String code)
        throws CountryNotFoundException {

        return service.getCountry(code);
    }
}

```

application.properties

```
server.port=8083
```

SpringLearnApplicationTests.java

```

package com.cognizant.springlearn;

import static org.junit.jupiter.api.Assertions.assertNotNull;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static
org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.AutoConfigureMockMvc;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.ResultActions;

import com.cognizant.springlearn.controller.CountryController;

@SpringBootTest
@AutoConfigureMockMvc
class SpringLearnApplicationTests {

    @Autowired
    private CountryController countryController;

```

```
@Autowired
private MockMvc mvc;

@Test
void contextLoads() {

    assertNotNull(countryController);
}

@Test
void testGetCountry()
    throws Exception {

    ResultActions actions =
        mvc.perform(get("/country"));

    actions.andExpect(status().isOk());

    actions.andExpect(
        jsonPath("$.code")
            .exists());

    actions.andExpect(
        jsonPath("$.code")
            .value("IN"));

    actions.andExpect(
        jsonPath("$.name")
            .exists());

    actions.andExpect(
        jsonPath("$.name")
            .value("India"));
}

@Test
void testGetCountryException()
    throws Exception {

    ResultActions actions =
```

```
        mvc.perform(get("/country/az"));

    actions.andExpect(
        status().isNotFound());

    actions.andExpect(
        status().reason(
            "Country not found"));
}
}
```

Test URLs

Method	URL	Output
GET	http://localhost:8083/hello	Hello World!!
GET	http://localhost:8083/country	India JSON
GET	http://localhost:8083/countries	All Countries
GET	http://localhost:8083/country/in	India
GET	http://localhost:8083/country/az	404 Error

- ▼ spring-learn-rest [boot] [devtools]
 - ▼ src/main/java
 - ▼ com.cognizant.springlearn
 - > Country.java
 - > SpringLearnApplication.java
 - ▼ com.cognizant.springlearn.controller
 - > CountryController.java
 - > HelloController.java
 - ▼ com.cognizant.springlearn.service
 - > CountryService.java
 - ▼ com.cognizant.springlearn.service.exception
 - > CountryNotFoundException.java
 - > src/main/resources
 - > src/test/java
 - > JRE System Library [JavaSE-21]
 - > Maven Dependencies
 - target/generated-sources/annotations
 - target/generated-test-sources/test-annotations
 - > src
 - > target
 - mvnw
 - mvnw.cmd









